

Donato Meoli

RESEARCH FELLOW · MATHEMATICAL OPTIMIZATION · SYMBOLIC AI · MACHINE LEARNING

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Summary

I came to numerical optimization from symbolic AI, and more precisely from Knowledge Representation and Reasoning (KR&R), i.e., expert systems, logic programming, constraint satisfaction and SAT. In my view, the two fields share one premise: the difficulty of a problem depends (often to a large extent) on how it is written. Hence, the useful move is frequently to rewrite it, either as a set of smaller problems coordinated by a master or in a different formalism for which an efficient algorithm is available; for instance, a planning problem can be stated as a Constraint Satisfaction Problem and, from there, as a propositional formula, at which point the machinery of logical reasoning applies to it.

On the numerical side, this means Lagrangian (Dantzig-Wolfe) Relaxation and Decomposition, Benders Decomposition, the Bundle Methods that solve the nondifferentiable duals they give rise to, and the parallel solution of the subproblems, which are typically many and independent by construction. Reformulation comes back inside the subproblems, as the recognition of a structure for which a specialized (often combinatorial) algorithm exists; since a decomposition method solves each of them many times with data that change rather little, much of my current work is on re-optimization, i.e., on reusing what has been computed for one instance in the solution of the next one, and on enlarging the class of subproblems that dynamic programming can solve efficiently.

I see machine learning, which connects the two sides, mainly as a way of learning heuristics, to be placed at the points where an exact algorithm makes a choice, and hence of making that algorithm faster: on the symbolic side, the branching of a SAT solver; on the numerical side, the branching of a branch-and-bound whose bounds come from a specialized relaxation (e.g., a Lagrangian one). The learned part makes the choice, while the guarantees still come from the exact method. Conversely, learning problems can be written as structured optimization models, to which the same decomposition and re-optimization machinery applies.

Publications

A. Frangioni, C. Gentile, **D. Meoli**. *Dynamic Programming for the Single-Unit Commitment Problem with Spinning Reserves and Reactive Power*. Submitted, 2026.

Reserves and reactive power are absorbed into the polynomial dynamic programming algorithms for the single-unit commitment problem at essentially no algorithmic cost, by a per-period partial minimization that folds them into an effective convex piecewise-quadratic production cost. On the instances tested, the two resulting solvers are about two orders of magnitude faster than a general-purpose MILP solver, even when the latter is given the best of the formulations considered.

D. Fioriti, A. Pampado, C. Scarpelli, G. Pasini, G. Lutzemberger, S. Barsali, D. Poli, **D. Meoli**, L. Mencarelli, A. Frangioni. *Enhancing Energy System Modelling with Advanced Mathematical Decomposition Techniques: Feasibility of Coupling SMS++ and PyPSA*. Proceedings of the 2025 IEEE International Conference on Environment and Electrical Engineering and 2025 IEEE Industrial and Commercial Power Systems Europe, pages 1–6, 2025. doi:10.1109/EEEIC/ICPSEurope64998.2025.11169291

Research Software

SMS++ • co-lead maintainer, with A. Frangioni, since 2021.

A framework for block-structured optimization, in which a model is a tree of nested blocks and an algorithm attaches to any of them; hence, a single instance can be solved monolithically, dualized, or split by scenario without rewriting the model.

aima-python • contributor since 2019.

Reference implementation of the algorithms in Russell & Norvig's *Artificial Intelligence: A Modern Approach*, of which I am the fourth-most-active contributor by commits. Contributed to SAT solving and constraint propagation, the translation of planning problems into search, constraint satisfaction and SAT, linear learners and support vector machines, and the algorithms of the fourth edition that were still missing.

OptiML • author, since 2020.

Optimization algorithms, from line search and Newton-type methods to proximal bundle and interior point, together with the learning models trained by them, which allows one to study a method against the model it serves.

Elsewhere. Earlier work in symbolic reasoning and knowledge representation: *NeuroSAT*, neuro-symbolic branching heuristics for satisfiability; *PESAD* and *BeerEX*, expert systems in logic programming and in rules with fuzzy reasoning; *WS4J* and *WNAffect*, semantic relatedness and affect over a lexical ontology.

Experience

University of Pisa

Dipartimento di Informatica, DESTeC

RESEARCH FELLOW

Dec. 2021 - Present

- Decomposition methods and solvers for large-scale optimization, and core development of SMS++, within the AUTENS, RESILIENT and Power-Max research projects.
- Applied to planning and scheduling under uncertainty, i.e., to models on which a general-purpose MILP solver typically does not close the instance in reasonable time.

Exprivia S.p.A.

Italy

DATA SCIENTIST

Sept. 2018 - Feb. 2019

- Learning models for predictive maintenance on multivariate time series, for image classification, and for real-time detection and counting.

Reports

Support Vector Machines

University of Pisa

A COMPARATIVE STUDY OF THE FORMULATIONS FROM AN OPTIMIZATION POINT OF VIEW

June 2021

Education

University of Pisa

Italy

MSC IN COMPUTER SCIENCE, ARTIFICIAL INTELLIGENCE

University of Bari “Aldo Moro”

Italy

BSC IN COMPUTER SCIENCE

Grade: 110/110 cum laude